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Impact of speech-stream segmentation on improving listening comprehension and linguistic self-confidence in primary Arabic and English language classrooms

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ABSTRACT

In a quasi-experimental mixed-methods study, 112 native Arabic-speaking primary school students in Saudi Arabia learning English as a foreign language were assessed using various measures to determine the potential impact of speech-stream segmentation on their Arabic and English listening comprehension and linguistic self-confidence. Pre- and post-test listening comprehension tests, questionnaires, and interviews in both languages were employed in this study. From the initial sample, 56 Arabic language students were randomly divided into two groups: an experimental group, who used speech-stream segmentation strategies in all listening activities, and a comparison group, who were taught the same lessons using traditional methods. Similarly, 56 English language students were randomly selected from the same school and divided into two groups with the same instructional design. The results demonstrated statistical significance in the groups using speech-stream segmentation to enhance their listening comprehension and linguistic self-confidence in both first and foreign languages. The results further indicate that, among speech-stream segmentation strategies, the L1 group relies more on phonological knowledge, allophonic variation, and prosodic cues, whereas the EFL group places greater emphasis on segmenting the speech stream, recognizing distributional patterns, and identifying word boundaries. Several factors impacted the effectiveness of speech-stream segmentation in improving listening comprehension and linguistic self-confidence; however, these factors differed between the two languages. Key areas were outlined for future research on speech-stream segmentation and listening comprehension.

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Speech-stream segmentation; listening comprehension; linguistic self-confidence; first and foreign languages; factors

Introduction

Interest in languages and teaching languages has a rich history, with the four language skills (i.e. reading, writing, listening, and speaking) serving as fundamental elements of instruction. Although public examinations generally assess students' proficiency by treating each language skill equally, the importance of listening is growing in our

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interconnected world. In particular, technological advances in global communication have increased the need for students to develop strong listening skills due to the similarly increased potential for misunderstanding (Rost 2024). Consequently, teaching listening comprehension has become essential (Miao, Kang, and Meng 2025).

Listening to new languages poses unique challenges for students due to the significant prosodic differences between languages (McAndrews 2023). Prosody, which encompasses the rhythm, stress, and intonation of speech, varies widely between languages, affecting how students perceive and process spoken language. For instance, in syllable-timed languages, such as French or Italian (Ordin and Polyanskaya 2014), each syllable is given equal time, which creates a different auditory experience compared to stress-timed languages, such as English, in addition to languages that can exhibit both syllable- and stress-timed characteristics, such as Arabic (Ashour 2017). This rhythmic disparity can make it particularly difficult for students to identify word boundaries and understand speech flows. Additionally, the distinct phoneme–grapheme correspondence in different languages can make speech flows difficult for listeners to segment (Han 2022). These challenges can contribute to heightened anxiety and feelings of demoralization among second-language (L2) students (Tsang 2022).

Much of the literature has concentrated on broad listening teaching strategies (Dalman and Plonsky 2025; Goh and Vandergrift 2021), identifying listening challenges (Jia and Hew 2023), exploring the relationship between linguistic knowledge and strategies in listening comprehension (Fung and Macaro 2021), examining the link between listening comprehension strategies and proficiency (Milliner and Dimoski 2024), and investigating the connection between listening anxiety and listening efficacy (Canaran et al. 2024). In contrast, there has been less focus on listening comprehension across two languages or strategies designed explicitly for segmenting speech. This study aims to fill this research gap in the literature by employing a mixed-methods experimental approach that specifically investigates the impact of speech-stream segmentation on improving listening comprehension and linguistic self-confidence among students of both Arabic as a first language (L1) and English as a foreign language (EFL). In addressing these areas, this study seeks to answer the following four research questions (RQs):

RQ1 What is the impact of speech-stream segmentation training on improving students' listening comprehension in L1 Arabic and EFL?

RQ2 What types of speech-stream segmentation strategies were implemented by L1 Arabic and EFL experimental groups of students?

RQ3 What is the impact of speech-stream segmentation training on improving students' linguistic self-confidence in L1 Arabic and EFL?

RQ4 What factors impact speech-stream segmentation on improving students' listening comprehension and linguistic self-confidence in L1 Arabic and EFL?

Theoretical framework

This section explores the theoretical foundations of speech-stream segmentation, followed by a literature review that critically examines the interplay between listening comprehension and linguistic self-confidence within L1 and L2 contexts.

Speech-stream segmentation

Several studies have established a strong association between speech-stream segmentation and language learning, with comprehensive literature reviews summarizing findings that assert the crucial role of this cognitive process (Charles et al. 2015; Tremblay et al. 2016; Yurovsky, Yu, and Smith 2012). For example, small-scale studies have demonstrated that recognizing the spaces between words, a process known as segmenting the speech stream, is essential for listeners to comprehend spoken language effectively (Al-Jarf 2025; Simpson 2017). However, this process can be challenging for students, particularly those who are still developing their language skills. While listeners may perceive gaps between words, this perception can be misleading. In reality, speech flows continuously, with pauses occurring at regular intervals throughout the discourse (Field 2010). As a result, the listener's cognitive processes must work to segment or 'chunk' this continuous stream into smaller, manageable units, such as words or clauses (Field 2010; Simpson 2017). Early studies have highlighted that one of the most frequently reported challenges faced by students is their inability to chunk speech streams effectively (Goh 2000; Goh and Taib 2006). Identifying word boundaries is a fundamental aspect of Field's (2013) framework, as this plays a vital role in enabling listeners to extract meaning from acoustic signals. Some listeners often struggle with this segmentation process, which leads them to focus on individual words (Graham 2006; Graham and Macaro 2008). To address these challenges, teachers must exercise caution when addressing their students' listening problems. It is essential to focus not only on expanding students' vocabulary but also on ensuring that students possess adequate phonological knowledge and the necessary skills to employ segmentation cues effectively (Simpson 2017).

The literature related to speech segmentation strategies has shown that auditory signals include various language-specific cues, including allophonic variations influenced by word and phrase boundaries, as well as distributional patterns, such as consonant phonotactics and vowel harmony, which necessitate linguistic experience and phonological generalization (White 2018). Prosodic cues require students to classify syllables as strong or weak and understand their distributional regularities for effective metrical segmentation (Endress and Hauser 2010). However, cues to word boundaries are not deterministic. For example, silent pauses can occur within words during disfluent speech, and listeners must generalize based on the relevant phonological categories and infer heuristics regarding cue reliability (Matzinger, Ritt, and Fitch 2021). For instance, while most English content words begin with strong syllables, relying solely on metrical segmentation can lead to misparsing (Content, Kearns, and Frauenfelder 2001). Additionally, the strength of acoustic–phonetic and prosodic cues varies with contexts, speech rates, the speaker's awareness of ambiguity, and the listeners' needs (Giovannone and Theodore 2021; Smith 2015). Dobrego, Konina, and Mauranen (2023) suggested that speech segmentation studies should consider speech when examining fluent speakers, regardless of their L1, to achieve a more comprehensive understanding of the processes and effects of speech segmentation.

Speech segmentation in Arabic faces unique challenges due to the language's complex phonetic, morphological, and orthographic features (El Gayar and Suen 2018). A primary difficulty arises from the wide variety of vowel sounds, consonant clusters, and allophonic variations in spoken Arabic, which hinder clear identification of phoneme boundaries

(Mait and Aboutabit 2025). Additionally, the cursive nature of Arabic script adds further complexity, as the shape of each letter changes depending on its position in a word (initial, medial, final, or isolated), complicating the connection between written and spoken forms. The root-and-pattern morphological system of Arabic presents additional challenges: words are formed by applying vocalic patterns to consonantal roots, leading to derivations that affect word boundaries and segmentation (Zerari 2019). These factors pose significant challenges for developing accurate Arabic speech recognition and segmentation systems. Research on Arabic speech segmentation highlights the importance of using phonological knowledge, including allophonic variations such as assimilation and elision, prosodic cues like stress and intonation, word boundary detection, and auditory perception (Chakhtouna, Sekkate, and Adib 2022; Mait and Aboutabit 2025).

Listening comprehension

In recent decades, studies related to listening comprehension have transformed the way in which listening is considered. This evolution in our understanding of listening comprehension has shifted from its simple definition, which viewed listening as merely the ability to understand text read aloud (Hogan, Adlof, and Alonzo 2014) to embracing a more complex perspective that emphasizes the importance of listening's cognitive and metacognitive components, which integrate spoken text elements with existing knowledge to foster a cohesive understanding (Holschuh and Lampi 2018). Literature reviews have shown that listening comprehension involves several components, including meaning, word recognition, syntax, vocabulary, and the relationships between words (Cutler, Dahan, and Van Donselaar 1997; Van Zeeland and Schmitt 2013). Listening comprehension requires the quick processing of meaning and linguistic skills due to the transient nature of spoken language (Andringa et al. 2012; Babayiğit and Shapiro 2020; Zoghlami 2015).

Emphasis has also been placed on understanding the differences in the mechanisms of listening comprehension between L1 and L2. For instance, Edele and Stanat (2016) tested the assumption that L1 listening comprehension significantly predicts L2 reading comprehension among language-minority students and found a consistent relationship between L1 listening and L2 reading comprehension in two language groups with differing language similarities, indicating that this effect is independent of language. Additionally, a correlation was identified among L1 English and EFL students, revealing that weaknesses in vocabulary and grammar are linked to underperformance in listening comprehension among EFL students (Babayiğit and Shapiro 2020). Furthermore, limited working memory capacity appears to play a significant role in L2 listening comprehension, serving as a general cognitive factor that operates independently of L2 linguistic knowledge, particularly among students with limited linguistic skills (Satori 2021). Students with higher listening proficiency achieved better comprehension scores and were more likely to perceive the L1 English accent as nativelike (Algana and Hardison 2024).

Although research on the impact of speech-stream segmentation on listening comprehension is not particularly extensive, its significance has been acknowledged in recent studies (Simpson 2017; Zoghlami 2023). A key finding highlights the critical role of low-level cognitive processes for novice and intermediate students, which are vital for enabling students to segment continuous speech into recognizable units (Graham

2017). Various ear-training activities can be used to enhance these students' listening skills (Graham 2017). For instance, activities that encourage students to anticipate the written forms of words they hear can significantly improve their ability to connect spoken language with written language. Additionally, combining reading with listening exercises allows students to reinforce their understanding of language structure and vocabulary in real-time (Zhang and Zhang 2022). Another effective approach is to draw attention to intonation patterns within the target language, as this can help students grasp the nuances of meaning and emotion conveyed through speech (Dylman, Champoux-Larsson, and Frances 2025). Nevertheless, gaps in knowledge and understanding persist regarding listening comprehension among L1 Arabic and L2 English listeners. Therefore, this study focuses on enhancing listening comprehension in these two languages.

Linguistic self-confidence

Linguistic self-confidence is a crucial aspect of language learning and communication, defined as the self-perception of language competence combined with low levels of anxiety (Tsang 2022). It plays a significant role in how individuals engage with a new language, impacting their ability to communicate effectively (Ghanbarpour 2016). Linguistic self-confidence can be categorized into two types: situation-specific self-confidence and general language self-confidence, both of which relate to students' cognitive and affective domains (Wu, Lu, and Yuan 2024). This self-confidence is linked to language-use anxiety, where anxiety levels and perceived competence influence the willingness to communicate in specific situations. Situation-specific self-confidence refers to individuals' perceptions of their language ability in particular contexts or situations, such as speaking in a classroom setting or during a casual conversation (Wu, Lu, and Yuan 2024). In contrast, general language self-confidence encompasses a broader sense of competence across various contexts, reflecting an overall perception of one's language skills (Ghafar 2023).

A substantial body of research has focused on linguistic self-confidence, which is a critical factor in language learning and communication (Saka and Merç 2021; Tsang 2022; Wu, Lu, and Yuan 2024). In particular, the intricate interrelationships between various elements, listening skills, personal interest in languages, and students' levels of linguistic self-confidence have been examined. These studies have highlighted the importance of listening as a foundational skill in language acquisition, suggesting that effective listening can enhance both interest in languages and the development of linguistic self-confidence (Canaran et al. 2024; Fathi, Derakhshan, and Torabi 2020; Tsang 2022). As students become more interested in languages, their self-confidence tends to increase, creating a positive feedback loop that can lead to improved proficiency (Fathi, Derakhshan, and Torabi 2020). Overall, these findings highlight the interconnection of these factors and underscore the need for teachers to create supportive and engaging learning experiences that foster both interest and confidence among language students.

Further studies have shown that linguistic self-confidence can be improved significantly by targeted strategies. For instance, Jamalifar and Salehi (2020) revealed that engaging students in planning task rehearsals not only enhanced their speaking abilities but also increased their willingness to communicate in an L2 by boosting their linguistic self-

confidence. This finding highlights the importance of structured practice in language-learning environments. Additionally, Su (2021) indicated that many students tend to perceive their speaking confidence as low, often accompanied by high levels of anxiety. In this context, students may employ communication strategies, such as message abandonment, nonverbal strategies, and social/affective strategies, to overcome their speaking challenges (Chou 2024). However, it is noteworthy that strategies such as thinking in the target language and focusing on accuracy are among the least used (Curry and Peeters 2025). From this perspective, teachers' pedagogical approach to improving linguistic self-confidence has become a crucial issue.

Methodology

Research design

This study used a mixed-methods intervention research methodology in which both quantitative and qualitative data were collected and analysed. This approach leveraged the strengths of both methods to provide a multidimensional perspective to address RQs (Creswell 2021). Qualitative data were contextualized by assessing the effectiveness and extent of study strategies. A key aim was to understand why speech segmentation strategies might improve listening comprehension and linguistic self-confidence among both L1 Arabic and L2 English students. Specifically, this study sought to explore whether different students attributed varying reasons to the success of speech-stream segmentation strategies, in which purely quantitative approaches may not capture these nuances. By examining the data from multiple perspectives, this study aimed to present a comprehensive view of how teaching speech segmentation strategies improves students' listening comprehension and linguistic self-confidence.

Procedure

This study employed a quasi-experimental mixed-methods design to explore the effect of speech-stream segmentation strategies in the target population. Participants were randomly selected and assigned to experimental and comparison groups (Creswell and Guetterman 2024). Both the experimental and comparison groups completed pre- and post-tests for listening comprehension and linguistic self-confidence questionnaires at three intervals (i.e. Times 1–3), with 18 lessons conducted between Time 1 and Time 3. Each lesson lasted 45 min and focused on listening skills. Following the pre-tests, post-tests, and questionnaires at Time 3, the students participated in post-lesson face-to-face interviews, which lasted for approximately 1 h and were audio recorded. Factors impacting speech-stream segmentation and their influence on improving listening comprehension and linguistic self-confidence among L1 Arabic and EFL students were explored.

Sampling participants

The target population comprised 112 10–11-year-old male students in Saudi Arabia who were studying L1 Arabic and EFL. The age of 11 years marks the final year of primary education, during which students must complete public exams. By the time they reach 10–11

years old, these primary school students have typically been studying both L1 Arabic and L2 English for nearly 6 years. The sampling frame was limited to a single school in southern Saudi Arabia that was randomly selected from other primary schools in the region. From this school, 56 students studying Arabic during the 2023–2024 academic year were randomly chosen and divided into an experimental group and a comparison group. Similarly, 56 students studying English in the same academic year were randomly selected from the same school and divided into an experimental group and a comparison group. Randomization is crucial for establishing causation in experimental approaches, as it helps address the numerous variables affecting participants and mitigates the differences within and between groups (Cohen, Manion, and Morrison 2018). Table 1 presents the participants' demographic characteristics in both the Arabic and English language experimental and comparison groups.

Four sets of students were included in the experimental and comparison groups for L1 Arabic and EFL, with each group comprising 28 male students aged 10–11 years. All four groups completed 18 lessons between the pre- and post-tests. The average pre-test listening comprehension scores for the L1 Arabic experiential and comparison groups were 11.03 (standard deviation [SD] = 1.79) and 11.29 (SD = 1.65), respectively. In contrast, the average pre-test listening comprehension scores for the EFL experiential and comparison groups were 9.60 (SD = 2.07) and 9.07 (SD = 1.82), respectively.

Instruments

The study maintained consistency by using identical instruments for both groups, including comparable listening comprehension tests, linguistic self-confidence questionnaires, and interviews. When exploring supportive or conflicting relationships between the two languages, it is vital to consider these interdependencies (Creswell and Guetterman 2024). A multi-step adaptation process was carried out to verify the instruments' suitability for the specific context. The final English versions were translated into Arabic by the researcher, a native Arabic speaker fluent in English with extensive experience in language learning. Subsequently, five university instructors and four linguists examined both versions to assess validity, confirming that each item accurately reflected its intended construct in both languages. Using different instruments for the two groups would have posed significant challenges in comparing results, as variations in

Table 1. Demographic characteristics in the experimental and comparison groups.

Language	L1 Arabic		EFL	
Group	Experimental	Comparison	Experimental	Comparison
Number of students	28	28	28	28
Gender	Male	Male	Male	Male
Age	10–11 years	10–11 years	10–11 years	10–11 years
Number of lessons between pre – and post-tests	18	18	18	18
Average pre-test listening comprehension score	11.03 (SD = 1.79)	11.29 (SD = 1.65)	9.60 (SD = 2.07)	9.07 (SD = 1.82)

Abbreviations: EFL, English as a foreign language; L1, first language; SD, standard deviation.

measurement accuracy, construct representation, and cultural relevance could emerge (Mackey and Gass 2015). These discrepancies might undermine the validity of conclusions regarding the relationship between the languages, as differences in instrument design could confound observed effects rather than reflect genuine linguistic interdependencies.

Listening comprehension tests

The listening comprehension tests consisted of 32 multiple-choice questions in which the participants were asked to provide a brief, open-ended written explanation justifying their chosen answers. These listening comprehension tests took approximately 45 min to complete, including both open- and closed-ended questions. The closed-ended questions were formulated based on a variety of listening comprehension studies (Hagen et al. 2022; Milliner and Dimoski 2024; Wakamoto and Rose 2021). The listening comprehension test was translated into both Arabic and English. The listening comprehension test items were refined for contextual relevance, and their validity and reliability were pilot-tested among 14 bilingual students. Additionally, four linguistics scholars and five education scholars with experience in language teaching and learning reviewed each item in both the Arabic and English tests to make the necessary editorial revisions. Following their feedback, the listening comprehension test items were improved, and the subscales and questions were modified (see Table 2).

Testing listening comprehension presents challenges for teachers, as it is often unclear whether errors arise from the students' lack of understanding of spoken language or from their difficulties with questions, as their reading and writing skills may potentially confound the results. To address these challenges, the comparison groups benefited from receiving traditional instruction as part of their regular learning. In contrast, the experimental groups focused on understanding the input itself using speech-stream segmentation strategies in in-class activities that did not resemble the tasks in the pre- or posttests (Cross and Vandergrift 2015). The underlying premise is that the students' ability to segment the speech stream is crucial for decoding aural input, leading to the hypothesis that speech-stream segmentation training is necessary before effectively teaching higher-level comprehension strategies (Goh and Vandergrift 2021). Although the pretest and posttest were identical in both languages, calibrated at the same level and with comparable cognitive demands on participants, the activities included in each required students to focus on reading questions and identifying speech segmentation strategies to aid in finding answers. These activities allowed the researcher to determine the most relevant strategies for L1 Arabic and EFL and to explore the similarities and differences between the two groups.

Table 2. Number of questions related to the five subscales in the listening comprehension test.

Scale	No. of questions
Word recognition	8
Vocabulary knowledge	7
Grammatical knowledge	5
Meaning	6
Metacognition	6
Total number of questions	32

Linguistic self-confidence questionnaires

The linguistic self-confidence questionnaire consisted of nine closed-ended statements that assessed the participants' linguistic self-confidence in relation to their listening progress throughout the study. Participants indicated their level of agreement with each statement on a 5-point Likert scale (1 = 'strongly disagree' and 5 = 'strongly agree'). The questionnaire took approximately 10 min to complete. The nine statements were formulated based on earlier linguistic self-confidence studies (Canaran et al. 2024; Curry and Peeters 2025; Su 2021; Tsang 2022). Both experiential groups completed a questionnaire in Arabic and English at each of the three time stages to provide insights into the participants' linguistic self-confidence over time. The questions from Time 1 were repeated at Times 2 and 3 alongside additional questions introduced at Times 2 and 3 to explore any improvements in the participants' linguistic self-confidence.

The questionnaire items were refined for contextual relevance in three steps. First, the validity and reliability of the linguistic self-confidence questionnaire items were pilot tested. Second, two psychologists and five education academics reviewed the questionnaire. Finally, the Cronbach's α reliability coefficient was calculated to assess the questionnaire's reliability and determine its internal consistency. The Cronbach's α reliability coefficient is 0.757. Teddlie and Tashakkori (2021) indicated that a Cronbach's α of $\geq$0.70 is necessary to demonstrate acceptable and satisfactory internal consistency.

Interviews

Alongside individual details, the quantitative data provided an overview of the participants' listening comprehension and linguistic self-confidence. The interviews' primary aim was to gain a deeper understanding of the factors that affect the success or failure of speech-stream segmentation training to address *RQ1* and to determine how these factors may impact improvements in students' listening comprehension and linguistic self-confidence. Twelve participants from both the L1 Arabic and EFL experimental groups were interviewed using a purposeful sampling method (Mills, Gay, and Gay 2015) to ensure that they represented diverse proficiencies in listening comprehension and linguistic self-confidence in both Arabic and English, as indicated by their pre- and post-test scores.

Intervention

Following the sequential steps outlined in Figure 1, students in both the L1 Arabic and EFL experimental groups were introduced to speech-stream segmentation strategies. The students focused on how words can blend together, creating the illusion that the final phoneme of one word is the initial phoneme of the next. A series of dictation exercises followed, ranging from two-word phrases to whole paragraphs, which were either read aloud by the teacher or played from recordings.

To enhance their ability to segment speech, the students reviewed the listening exercises with transcripts in hand. Later in the intervention, the teacher/researcher provided transcripts with deliberate errors, including minimal pairs and changes to key words, such as adverbs of time, to emphasize that the crucial words for understanding texts are not always nouns and verbs. The students were tasked with identifying and correcting these transcript errors. In addition, some transcripts contained gaps for students to fill in.

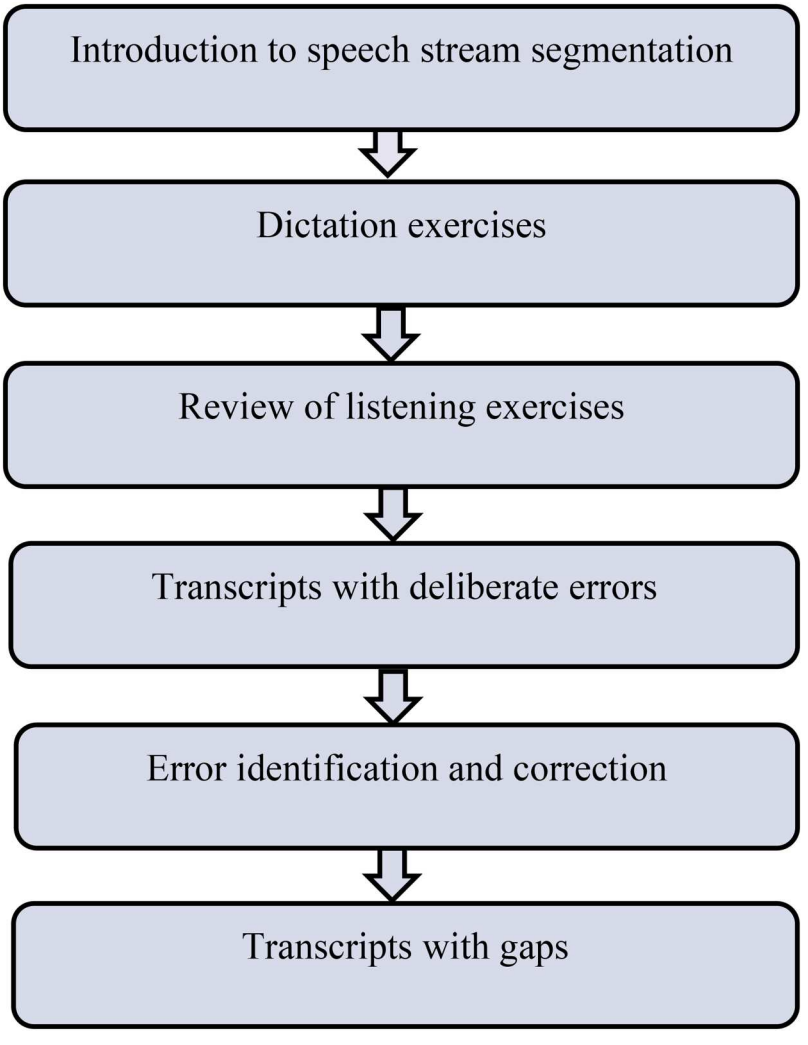


Figure 1. Sequential steps used in the speech-stream segmentation intervention.

Data analysis

Multiple sources were used to collect data about the participating students, and both quantitative and qualitative analyses were conducted (see Table 3).

The SPSS statistical package (v. 29; IBM SPSS, Armonk, NY, USA) was used to analyse the quantitative results for *RQ1*, *RQ2,* and *RQ3*. Descriptive statistics and repeated-measures analyses of variance (ANOVAs) were conducted using pre- and post-test data and the questionnaire responses to examine the impact of the speech-stream segmentation intervention on the experimental groups with respect to the comparison groups. The focus was on improvements in listening comprehension and linguistic self-confidence across three time points before and after the speech-stream segmentation training activities.

To address *RQ4*, thematic analysis was performed using NVivo software (Lumivero, Burlington, MA, USA), which is designed for qualitative data analysis (Allsop et al. 2022).

Table 3. Data analysis methods and sources that shaped the RQs.

RQs	Data sources that informed the RQs	Data analysis
What is the impact of speech-stream segmentation training on improving listening comprehension in L1 Arabic and EFL?	Pre- and post-tests	Descriptive statistics Repeated-measures ANOVA
What is the impact of speech-stream segmentation training on improving linguistic self-confidence in L1 Arabic and EFL?	Questionnaires	Descriptive statistics
What factors impact speech-stream segmentation on improving listening comprehension and linguistic self-confidence in L1 Arabic and EFL?	Interviews	Codes and themes Similar and different themes and categories Constant comparative analysis method

Abbreviations: ANOVA, analyses of variance; EFL, English as a foreign language; L1, first language; RQs, research questions.

Interview recordings were examined to identify the factors influencing speech-stream segmentation and its effects on improving listening comprehension and linguistic self-confidence. The constant comparative analysis method, which includes inductive qualitative analysis, was used to identify similarities and differences, as well as to explore codes, themes, and categories (Strauss 1987; Thompson Burdine, Thorne, and Sandhu 2021).

The study design incorporated ethical considerations to gain access to sites and participants. After approval from the Ethics Committee approved the protocol, permission was obtained from the Ministry of Education's gatekeeper and the headteachers of the selected schools. To ensure voluntary participation and informed consent, teachers, parents and students signed consent forms at the beginning of the study. Participants were informed about the objectives of the study, the types of data to be collected, and their responsibilities. Anonymity was preserved through coding, with real names removed before analysis and replaced with pseudonyms. Confidentiality was maintained throughout audio transcription, data analysis, and the presentation of results.

Results

The experimental results are structured around the three RQs. First, the listening comprehension of the L1 Arabic and EFL experimental and comparison groups was analysed to assess the effectiveness of the speech-stream segmentation intervention. Second, the questionnaire data were examined to evaluate the impact of the intervention on the experimental groups' linguistic self-confidence. Finally, the semi-structured interview findings were presented to explore the factors influencing the quantitative results.

Impact of speech-stream segmentation on improving the listening comprehension of L1 Arabic and EFL students

The results of the pre- and post-tests for both groups are presented in Table 4 and illustrated graphically in Figure 2. These findings indicate a significant difference in improving listening comprehension between the L1 Arabic and EFL experimental and comparison groups over time, favoring the experimental groups. For the L1 Arabic groups, the results of repeated-measures ANOVA revealed that the mean scores differed significantly

Table 4. Results of listening comprehension tests for L1 Arabic and EFL students.

Language	Test	Group	*N*	*M*	SD	*F*	*P*	η^2
L1 Arabic	Pre-test 1	Experimental	28	11.03	1.79	269.982	<.001	.831
		Comparison	28	11.29	1.65			
	Post-test 2 (after 6 weeks)	Experimental	28	27.30	6.91			
		Comparison	28	11.22	1.84			
	Post-test 3 (after 12 weeks)	Experimental	28	36.36	5.39			
		Comparison	29	11.40	1.67			
EFL	Pre-test 1	Experimental	28	9.60	2.07	282.902	<.001	.840
		Comparison	28	9.07	1.82			
	Post-test 2 (after 6 weeks)	Experimental	28	27.42	7.32			
		Comparison	28	10.64	1.85			
	Post-test 3 (after 12 weeks)	Experimental	28	37.21	5.56			
		Comparison	28	10.89	1.83			

Abbreviations: EFL, English as a foreign language; L1, first language; M, mean; SD, standard deviation.

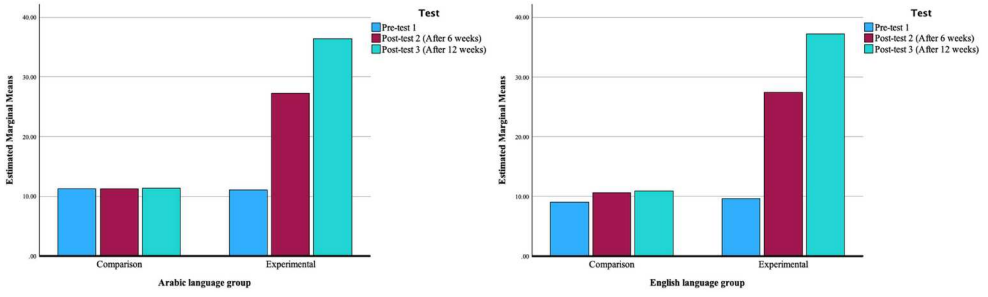


Figure 2. Results of listening comprehension tests for L1 Arabic and EFL students.

across the three time points ($F = 269.982$, $p < .001$). At the pre-test, the mean score for the L1 Arabic experimental group was 11.22 (SD = 1.79), while the mean score for the L1 Arabic comparison group was 11.29 (SD = 1.65). At Time 2, the mean post-test score for the experimental group increased significantly to 27.30 (SD = 6.91), while the comparison group had a mean score of 11.22 (SD = 1.84). By Time 3, the mean post-test score for the experimental group further increased to 36.36 (SD = 5.39), demonstrating a continuous improvement, while the comparison group had a mean score of 11.40 (SD = 1.67). The effect size estimate for speech-stream segmentation in improving listening comprehension among L1 Arabic primary school students was $\eta^2 = .831$.

Similarly, for the EFL experimental and comparison groups, a repeated-measures ANOVA indicated significant differences in mean scores across the three time points ($F = 282.902$, $p < .001$), with an effect size estimate of $\eta^2 = .840$ for speech-stream segmentation in improving listening comprehension among EFL students.

At the pre-test, the mean score for the EFL experimental group was 9.60 (SD = 2.07), compared to 9.07 (SD = 1.82) for the EFL comparison group. By Time 2, the mean post-test score for the experimental group increased significantly to 27.42 (SD = 7.32), while the comparison group maintained a mean score of 10.64 (SD = 1.85). By Time 3, the experimental group's mean post-test score increased further to 37.21 (SD = 5.56), indicating a consistent improvement, whereas the comparison group had a mean score of 10.89 (SD = 1.83).

Types of speech-stream segmentation strategies among L1 Arabic and EFL experimental groups

As shown in Table 5 and Figure 3, speech-stream segmentation strategies used by both L1 Arabic and EFL experimental groups–applied across three time points–include: detecting the spaces between words, segmenting the speech-stream into smaller units, using allophonic variations, recognizing distributional patterns, applying phonological knowledge, and uilizing prosodic cues. However, closer examination revealed that the L1 Arabic experimental groups relied more on applying phonological knowledge, using allophonic variations, and uilizing prosodic cues, whereas the EFL experimental groups concentrated more on segmenting the speech-stream into smaller units, recognizing distributional patterns, and detecting the spaces between words.

Impact of speech-stream segmentation on improving the linguistic self-Confidence of L1 Arabic and EFL students

Table 6 presents the descriptive statistics for the questionnaire, which was administered exclusively to both L1 Arabic and EFL experimental groups at Times 1–3. The first five questions were asked at Time 1 and repeated at Times 2 and 3, when four additional statements were included, resulting in a total of nine statements assessed at these time points.

Table 5. Speech-stream segmentation strategies among L1 Arabic and EFL experimental groups.

	L1 Arabic			EFL		
Speech-Stream Segmentation Strategies	T1 % (*n*)	T2 % (*n*)	T3 % (*n*)	T1 % (*n*)	T2 % (*n*)	T3 % (*n*)
1. Detecting the spaces between words	9.09 (3)	16.75 (32)	6.11 (63)	15.38 (4)	14.53 (39)	18.69 (118)
2. Segmenting speech-stream into smaller units	15.15 (5)	14.65 (28)	9.36 (49)	34.62 (9)	38.29 (103)	45.57 (287)
3. Using allophonic variations	21.21 (7)	23.56 (45)	18.73 (98)	7.69 (2)	8.92 (24)	4.44 (28)
4. Recognizing distributional patterns	18.18 (6)	9.42 (18)	6.50 (34)	15.38 (4)	14.87 (40)	19.01 (120)
5. Applying phonological knowledge.	24.24(8)	25.65 (49)	36.32 (190)	23.08 (6)	16.73 (45)	7.45 (47)
6. Utilizing prosodic cues	12.12 (4)	9.94 (19)	17.01 (89)	3.85 (1)	6.69 (18)	4.91 (31)
Total	100 (33)	100 (191)	100 (523)	100 (26)	100 (269)	100 (631)

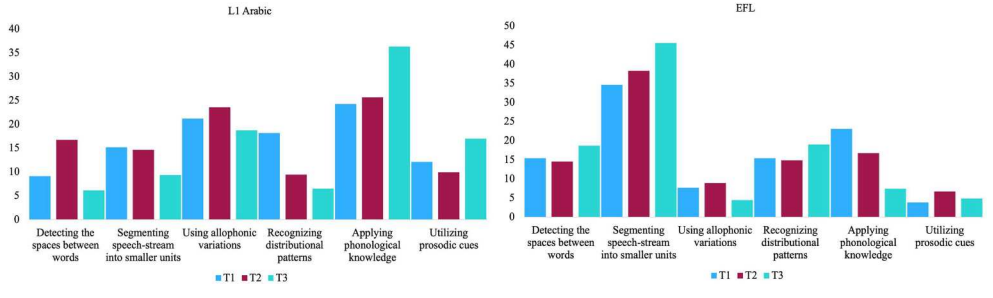


Figure 3. Speech-Stream Segmentation Strategies among L1 Arabic and EFL experimental groups.

Table 6. Results of linguistic self-confidence for L1 Arabic and EFL students.

	Time 1		Time 2		Time 3	
Statement	L1 Arabic (n = 28) M (SD)	EFL (n = 28) M (SD)	L1 Arabic (n = 28) M (SD)	EFL (n = 28) M (SD)	L1 Arabic (n = 28) M (SD)	EFL (n = 28) M (SD)
I feel apprehensive when I must summarize what I've understood from the topic.	4.00 (0.26)	4.07 (0.57)	1.85 (0.93)	1.85 (0.84)	1.53 (0.83)	1.57 (0.63)
I feel capable of participating in language learning activities.	1.94 (0.35)	1.92 (0.37)	4.14 (0.44)	4.28 (0.46)	4.42 (0.50)	4.50 (0.50)
I am willing to engage in a language comprehension task.	1.89 (0.90)	1.85 (0.93)	4.35 (0.48)	4.50 (0.50)	4.71 (0.46)	4.78 (0.41)
I feel I have the competence to interact with a given topic.	1.91 (0.85)	1.71 (0.89)	4.32 (0.47)	4.42 (0.50)	4.78 (0.41)	4.75 (0.44)
I feel comfortable speaking in the front of the whole class.	1.96 (0.88)	1.85 (0.84)	4.35 (0.48)	4.42 (0.50)	4.57 (0.50)	4.64 (0.48)
The speech-stream segmentation intervention generally improved my self-confidence.	–	–	4.42 (0.49)	4.28 (0.46)	4.75 (0.64)	4.89 (0.31)
The speech-stream segmentation intervention boosted my confidence in learning language.	–	–	4.41 (0.49)	4.57 (0.50)	4.89 (0.31)	4.78 (0.41)
The speech-stream segmentation intervention altered my perspective on language learning.	–	–	4.42 (0.49)	4.53 (0.50)	4.89 (0.31)	4.92 (0.26)
The exam would have been more challenging if I had not participated in the speech-stream segmentation intervention.	–	–	4.01 (0.40)	4.71 (0.46)	4.85 (0.35)	4.82 (0.39)

Abbreviations: EFL, English as a foreign language; L1, first language.

As shown in Table 6, the negative statements decreased from a high to a low level of agreement, indicating that the students perceived improvements in their linguistic self-confidence. For example, in response to the negative statement, 'I feel apprehensive when I must summarize what I've understood from the topic', the mean response decreased from 4.00 (SD = 0.26) and 4.07 (SD = 0.57) at Time 1 in the L1 Arabic and EFL experimental groups, respectively, to 1.85 (SD = 0.93) and 1.53 (SD = 0.84) at Time 2 in the L1 Arabic and EFL experimental groups, respectively. This trend continued to 1.53 (SD = 0.83) and 1.57 (SD = 0.63) at Time 3 in the L1 Arabic and EFL experimental groups, respectively.

In contrast, the positive statements increased from a low to a high level of agreement, indicating that the students perceived improvements in their linguistic self-confidence. For instance, in response to the positive statement, 'I feel I have the competence to interact with a given topic', the mean response increased from 1.91 (SD = 0.85) and 1.71 (SD = 0.89) at Time 1 in the L1 Arabic and EFL experimental groups at Time 1, respectively, to 4.32 (SD = 0.47) and 4.42 (SD = 0.50) at Time 2 in the L1 Arabic and EFL experimental groups, respectively, to 4.78 (SD = 0.41) and 4.75 (SD = 0.44) at Time 3 in the L1 Arabic and EFL experimental groups, respectively. Moreover, the students' perceived responses to statements 6–9 continued to increase to a high level of agreement following their involvement in the speech-stream segmentation intervention.

Factors impacting speech-stream segmentation

Seven factors influencing speech-stream segmentation were identified from the interview data, which affected the impact of the speech-stream segmentation intervention differently among the L1 Arabic and EFL students (see Table 7).

Table 7. Factors impacting the speech-stream segmentation intervention.

Group	Factors	
L1 Arabic	• Speech speed	• Variation in speech sounds
	• Acoustic–phonetic consistency	• Lexical knowledge
EFL	• Variation in syllable length	• Word recognition
	• Language exposure	• Cognitive load and recall ability

Abbreviations: EFL, English as a foreign language; L1, first language.

In the L1 Arabic experimental group, several factors impacted the speech-stream segmentation intervention, including speech speed, acoustic–phonetic consistency, variation in speech sounds, and lexical knowledge. Eight students described the speed of speech in Arabic as challenging. For example, Student B said, 'Although Arabic is my first language, I occasionally struggle to follow speakers who talk rapidly. I'm trying to catch what they're saying, but their speed makes it hard for me to understand the words'. Similarly, Student R said, 'I used to not understand a word or phrase because some speakers talk quickly; however, when I pay attention to the gaps between words, like what we have in this intervention, I can grasp their meaning, and I don't miss everything else'.

Students D, Y, and T indicated that acoustic–phonetic consistency helps them to better understand spoken text. As Student D stated, 'When the sounds match the letters, it makes it so much easier for me to follow along and grasp what's being said'. Students B, C, E, G, and K observed that variations in speech sounds frequently result in a loss of concentration and negatively impact their ability to listen. For example, Student G said, 'When the speech sounds change unexpectedly, it becomes hard for me to focus and understand what's being said'.

Students F, J, L, and Z emphasized that they sometimes struggle to create a mental representation of what they hear due to their insufficient lexical knowledge. Student F stated, 'I often find it difficult to visualize what I'm listening to because I don't know enough words'. Student J remarked, 'When I hear new terms, it's hard for me to connect them to what I already know'. Student L added, 'Without a strong vocabulary, I feel lost when trying to picture concepts in my mind'. Finally, Student Z expressed, 'I wish I could understand more words; it would help me create clearer images of what I hear'.

In contrast, speech-stream segmentation in English is influenced by variations in syllable length, language exposure, word recognition, cognitive load, and recall ability. Seven students described variations in the length of English syllables as challenging to understand. For example, Student V stated, 'I often struggle with words that have varying syllable lengths; it makes it hard for me to keep up with conversations'. Student Y remarked, 'When I hear English native speakers, the way they stretch or shorten syllables confuses me'. Student M also indicated that 'Understanding syllable length variation is crucial for me; it sometimes affects my understanding of what I'm listening to'. Student L added, 'In some exercises, especially at the beginning of this intervention, the variation in syllable length can be tricky; sometimes I miss the meaning'.

Five students, V, O, E, N, and W, indicated that their lack of language exposure significantly affected their speech-stream segmentation in EFL activities. Student V stated, 'Without enough exposure to English, I find it hard to break down sentences into

understandable parts'. Student O remarked, 'I often miss important words because I haven't heard them used in context enough'. Student E added, 'When I don't hear English spoken often, I struggle to separate words in a fast speech stream'. Finally, Student W expressed, 'I feel lost in discussions sometimes; if I had more practice, I could segment speech better'.

Students Y, L, and W indicated that word recognition can affect their EFL speech-stream segmentation. Student Y stated, 'I often struggle to recognize words quickly, which makes it hard for me to keep up with conversations'. Student L expressed, 'Word recognition is a big challenge for me; sometimes I know the meaning but can't identify the word when I hear it'. Student W added, 'I find that I miss out on key information because I can't recognize words quickly enough in spoken English'. Students J and E noted that their cognitive load and recall abilities influenced how they segmented English speech streams. In particular, Student J said, 'I've noticed that when my mental workload is high, it becomes much harder for me to pick out individual words in a fast speech stream'. Student E said, 'I agree! When I struggle to recall information, it feels like the speech just blurs together'.

Discussion

This study contributes to the literature by (a) investigating the impact of speech-stream segmentation on improving listening comprehension and linguistic self-confidence in L1 Arabic and EFL students and (b) identifying the factors affecting speech-stream segmentation on improving listening comprehension and linguistic self-confidence through a mixed-methods intervention methodology that combines both quantitative and qualitative results. In particular, the quantitative results indicate that speech-stream segmentation significantly improves listening comprehension among both L1 Arabic and EFL students. These findings indicate that the students who used speech-stream segmentation in the experimental groups demonstrated improved listening comprehension. This implies that their ability to identify individual words within a speech stream while listening activated their vocabulary knowledge, which may have been partially facilitated by their enhanced ability to visualize the orthographic forms of the words they were hearing (Field 2010). This result aligns with recent research in which listening comprehension was assessed following exposure to texts presented in various modalities (Clinton-Lisell 2022; Hui 2024; Rogowsky, Calhoun, and Tallal 2016). This result also suggests that speech-stream segmentation is not only effective for L2 or foreign language students (Simpson 2017; Tremblay et al. 2016), but also appears to be important and valuable when applied to L1 listening comprehension.

The quantitative data further indicate that both L1 Arabic and EFL experimental groups, evaluated at three time points, use a range of speech-stream segmentation strategies, including detecting the spaces between words, segmenting the speech-stream into smaller units, using allophonic variations, recognizing distributional patterns, applying phonological knowledge, and uilizing prosodic cues. This pattern suggests that strategies developed in L1 can be effectively transferred to L2 learning. Additionally, it underscores the important role of teachers in raising learners' awareness of these strategies in both their first and second languages. An in-depth analysis shows distinct differences between the groups. The L1 Arabic group relies more heavily on phonological knowledge,

allophonic variation, and prosodic cues, which reflects their greater familiarity with the sounds and rhythm of their native language. In contrast, the EFL group places greater emphasis on segmenting the speech stream, recognizing distributional patterns, and identifying word boundaries, likely to compensate for reduced lexical familiarity and contextual support in the second language.

Additionally, the quantitative results indicate that speech-stream segmentation significantly improved linguistic self-confidence among both L1 Arabic and EFL students. The ratings for linguistic self-confidence among the participants in both experimental groups demonstrated a statistically significant increase from Time 1 to Time 2 and again to Time 3, suggesting a correlation between their notable improvement in listening comprehension and enhanced linguistic self-confidence. This improvement in listening comprehension empowered these students to take on linguistic challenges, which

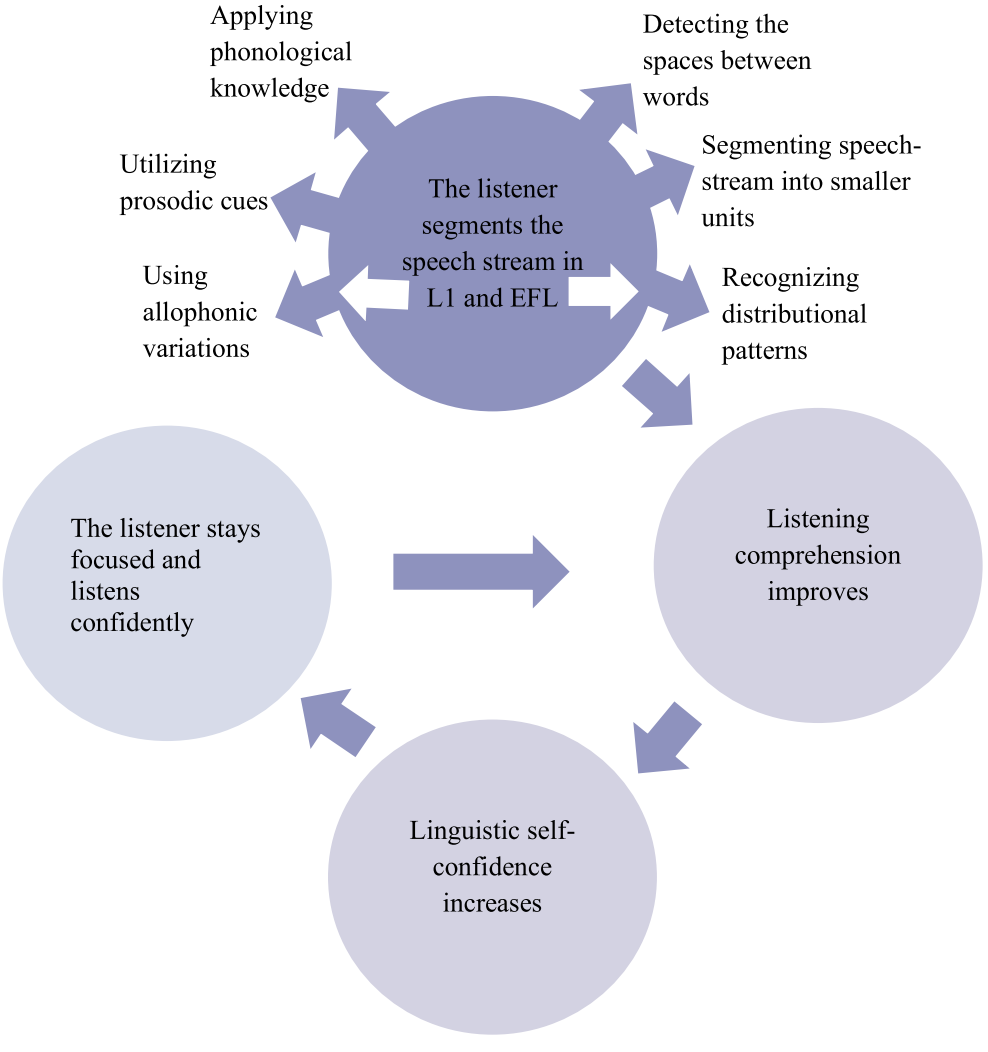


Figure 4. The impact of speech-stream segmentation on improving listening comprehension and linguistic self-confidence, along with speech-stream segmentation strategies.

strengthened their overall self-confidence. This aligns with the findings of Milliner and Dimoski (2024), who concluded that L2 listening fostered more positive attitudes among students. Therefore, the impact of speech-stream segmentation on improving students' listening comprehension in L1 Arabic and EFL (R1), the types of speech-stream segmentation strategies used by L1 Arabic and EFL (R2), and linguistic self-confidence (R3) can be summarized with a virtuous circle as illustrated in Figure 4.

The qualitative results (R4) indicate that several factors impact the effectiveness of speech-stream segmentation in improving L1 Arabic and EFL listening comprehension and linguistic self-confidence. However, it is important to note that these factors differ between the two languages. In the case of L1 Arabic, key factors impacting speech-stream segmentation include speech speed, acoustic–phonetic consistency, variation in speech sounds, and the listener's lexical knowledge. Conversely, in EFL, speech-stream segmentation is influenced by various factors, such as variations in syllable length, language exposure, word recognition, cognitive load, and recall ability. This distinction between L1 Arabic and EFL suggests that the cognitive processes involved in speech-stream segmentation may vary based on the linguistic characteristics of each language. Understanding these differences is crucial for effectively using speech-stream segmentation to enhance listening skills and linguistic self-confidence among language students, as the factors influencing this process in one language may not apply to another. Furthermore, linguistic research supports the idea that the core structural constituents of a given language should encompass both phonology and morphology (Kou et al. 2024; Zhang et al. 2023). This highlights the importance of considering these aspects when examining speech-stream segmentation across different languages.

Conclusion

This study employed a mixed-methods intervention methodology with a sample of 112 randomly chosen students from the same school, from which 56 L1 Arabic students and 56 EFL students were divided into experimental and comparison groups. The results indicated that explicit training in speech-stream segmentation plays a crucial role in improving both L1 Arabic and EFL students' listening comprehension and boosts their linguistic self-confidence.

Given these findings, it may be advantageous to implement a diverse range of listening strategies that include speech-stream segmentation techniques. This approach could cater to different learning styles and needs. As discussed in the literature review, the prosody of both Arabic and English is distinct and presents specific challenges for listeners of each language. Therefore, the study findings should not be generalized to students of other languages or to speakers of different languages who are learning Arabic or English.

Future research is necessary not only to replicate this study but also to conduct partial replications by varying the language combinations and exploring the impact of speech-stream segmentation strategies on the participants' other language skills. Furthermore, while this study was not explicitly designed to investigate the relationship between listening comprehension and linguistic self-confidence, it does present findings indicating improvements in both domains. Further research is warranted to examine the underlying causes and correlations between these two factors.

The qualitative results indicate that several factors impact the effectiveness of speech-stream segmentation strategies in improving Arabic and English speakers' listening comprehension and linguistic self-confidence. However, this study was a small-scale, school-based investigation involving only 112 students, who were divided into L1 Arabic and EFL groups with 56 students each and then subsequently divided again into experimental and comparison groups with 28 students each. More in-depth questionnaires with a larger number of participants using exploratory factor analyses might help to explore the typology of factors affecting the effectiveness of speech-stream segmentation strategies.

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Data availability statement

The data supporting this study's findings are available from the corresponding author upon reasonable request.

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